Does Length Buy Quality?

Runtime and Audience Reception in TMDb Film Data, 2000-2024

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1 Executive summary

This report asks a question a commissioning team has to answer every time it approves a project: **how long should a film be?** It uses 570,736 films released between 2000 and 2024, drawn from The Movie Database (TMDb), of which 18,329 carry both a plausible runtime and an audience score resting on enough votes to mean something.

Two consecutive CRISP-DM cycles were carried out. Cycle 1 asked whether runtime relates to audience score across the catalogue. It found that it does, but **not in the direction or the shape expected**: the relationship is not a straight line but a U, with the worst-rated films sitting in the 80 to 99 minute band and scores climbing steadily from there to over 7.0 for films of 150 minutes or more. The gap between the best and worst band is 1.03 rating points, comfortably larger than the threshold set in advance.

The obvious objection is that this measures genre rather than length, since documentaries and animated films are short while epics are long. Cycle 2 tested that directly by computing the runtime-rating relationship separately **within** each genre. It survives: 18 of 19 genres show a positive association, with a median of 0.25. But its strength varies sharply, and the pattern is interpretable — length tracks quality in narrative, plot-driven genres and barely at all in Family, Animation and Fantasy.

The recommendation that follows is not "make longer films." It is that runtime expectations should be set per genre rather than as a house rule, and that the 90-minute default which dominates the catalogue is the worst-performing length in it.

2 Introduction

2.1 The stakeholder and the business question

The stakeholder is the **commissioning team of a streaming platform**. They decide which projects to fund and, in doing so, set expectations about length: a 90-minute target is a common default, on the reasoning that shorter films are cheaper to produce and easier to schedule. That default is rarely tested.

The business question is whether it costs them anything. If longer films are systematically better received, a house rule that caps length is quietly suppressing quality; if they are not, the rule is free and should stay.

Translated into a data question:

> Across films released between 2000 and 2024, how does audience score vary with runtime — and does any relationship hold once genre is accounted for?

The framing is deliberately modest. Nothing in this data can show that *making* a film longer would improve it; films are made at the length their material requires. The claim available is one of association, and that is the only claim made.

2.2 Data description

The analysis uses 25 yearly extracts from TMDb, one per release year from 2000 to 2024, each with the same 24 columns. The variables this investigation depends on are:

- `runtime` — length in minutes, the explanatory variable
- `vote_average` — mean audience score out of 10, the outcome
- `vote_count` — how many votes that average rests on, used to decide whether the average is meaningful at all
- `genres` — a single comma-separated string per film, used in Cycle 2
- `release_date`, `status`, `adult`, `id` — used for scoping and de-duplication

The remaining columns (`budget`, `revenue`, `overview`, `tagline`, `homepage`, `imdb_id`, `poster_path` and others) are not used. `budget` and `revenue` were considered and rejected: both are recorded for only a small minority of titles, and a runtime question does not require them.

2.3 Tools and reproducibility

Tool	Role in this project
`ProjectTemplate`	Directory structure; automatic loading of the 25 yearly CSVs, the libraries, the `lib/` helpers and the cached objects
`dplyr`	All cleaning, filtering and aggregation, written as scripts in `munge/` so no manual data editing occurs anywhere
`ggplot2`	All figures, using the shared theme defined in `lib/helpers.R`
`R Markdown`	This report, knitted from the cached objects so the prose and the figures cannot drift apart
`Git`	Version control; the full commit history is in `GitLog.txt`
`renv`	Package versions fixed in `renv.lock`

Every number quoted in the text is inserted by inline R code rather than typed by hand. Re-running the project regenerates the prose as well as the plots, so a change to a threshold in `munge/` propagates automatically to every sentence that depends on it.

3 Cycle 1: runtime and audience score across the catalogue

3.1 Business understanding

The commissioning team's opening question is whether their 90-minute default costs them anything in audience reception.

Success criteria. This cycle will be considered to have answered its question if it can:

1. quantify the direction and size of any association between runtime and audience score across the whole catalogue;

2. establish whether that association is monotonic — that is, whether every step up in length moves the score the same way; and
3. treat a difference of at least **0.25 rating points** between the best- and worst-performing runtime bands as material. That threshold is roughly a third of the standard deviation of scores in the analysis subset, so anything smaller would be indistinguishable from noise at the level a commissioning decision operates.

Stating the threshold in advance matters: it is what makes the result falsifiable rather than a description of whatever the data happened to show.

3.2 Data understanding

Stacking the 25 yearly files gives 670,208 raw records. Table 1 shows the effect of each scoping and cleaning step.

Table 1: Effect of each scoping and cleaning step on the record count.

Cleaning step	Records remaining	Removed
Rows in the 25 raw files	670,208	-
with a valid TMDb id	670,208	0
after de-duplicating on id	670,072	136
with status 'Released'	661,569	8,503
excluding adult titles	570,736	90,833
with release year in 2000-2024	570,736	0

Table 2 gives the surface properties of the cleaned table for the variables this investigation uses: the type of each, how complete it is, and how many distinct values it takes.

Table 2: Surface properties of the variables used, across the 570,736 films in the cleaned table.

Variable	Type	Complete	Distinct values
id	integer	100.0%	570,736
title	character	100.0%	497,229
release_year	integer	100.0%	25
original_language	character	100.0%	166
runtime	numeric	78.0%	659
runtime_band	factor	78.0%	5
vote_average	numeric	35.5%	4,720
vote_count	integer	100.0%	3,176
genres	character	72.6%	9,887
primary_genre	character	72.6%	19

Four data quality issues emerged, each handled explicitly rather than silently, and each with a consequence for the analysis:

1. **Sentinel zeros.** `runtime` is recorded as `0` where the value is unknown. Treating those as genuine would imply thousands of films of zero length and would drag every mean downward. They are converted to `NA` in `munge/02-B-clean.R`. The same treatment is applied to `budget` and `revenue`.
2. **Ratings resting on almost no votes.** TMDb publishes a `vote_average` however few people voted, so a film with three votes can show 10.0. These are not measurements of audience opinion, and they produce the artificial spikes at whole and half numbers visible in Figure 2. **Consequence:** any analysis of scores without a vote threshold would be dominated by them, so a minimum of 50 votes is imposed and its effect is shown rather than assumed.
3. **Implausible runtimes.** The catalogue contains entries of a few minutes (trailers, music videos, data-entry errors) and of many hours (box sets catalogued as single titles). **Consequence:** both would distort a mean, so the analysis is restricted to 40 to 240 minutes.
4. **Inconsistent column types across files.** Because the yearly files are read independently, a column blank in one year is inferred as logical and as character elsewhere. `munge/01-A-combine.R` coerces everything to character before stacking and converts types once, centrally.

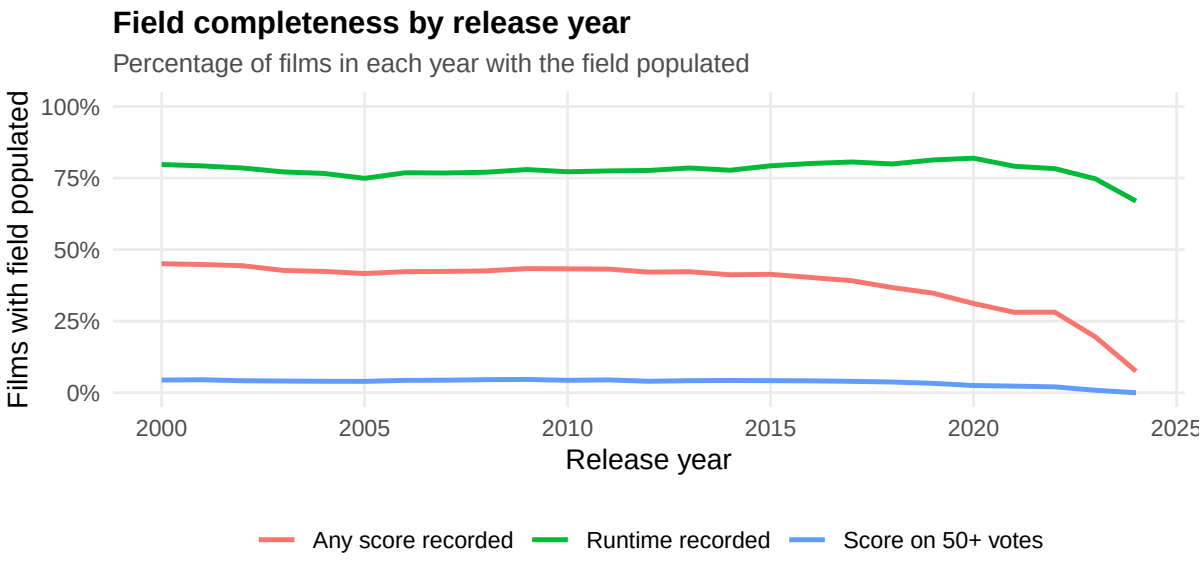


Figure 1: Completeness of the three fields this investigation depends on, by release year. Runtime and a raw score are recorded for most titles, but the share whose score rests on at least 50 votes is far smaller and falls steadily over the period.

Figure 1 shows why the vote threshold is the binding constraint rather than runtime. Runtime is recorded for the great majority of titles throughout, and so is some score; what is scarce, and getting scarcer, is a score that rests on enough votes to be worth reading.

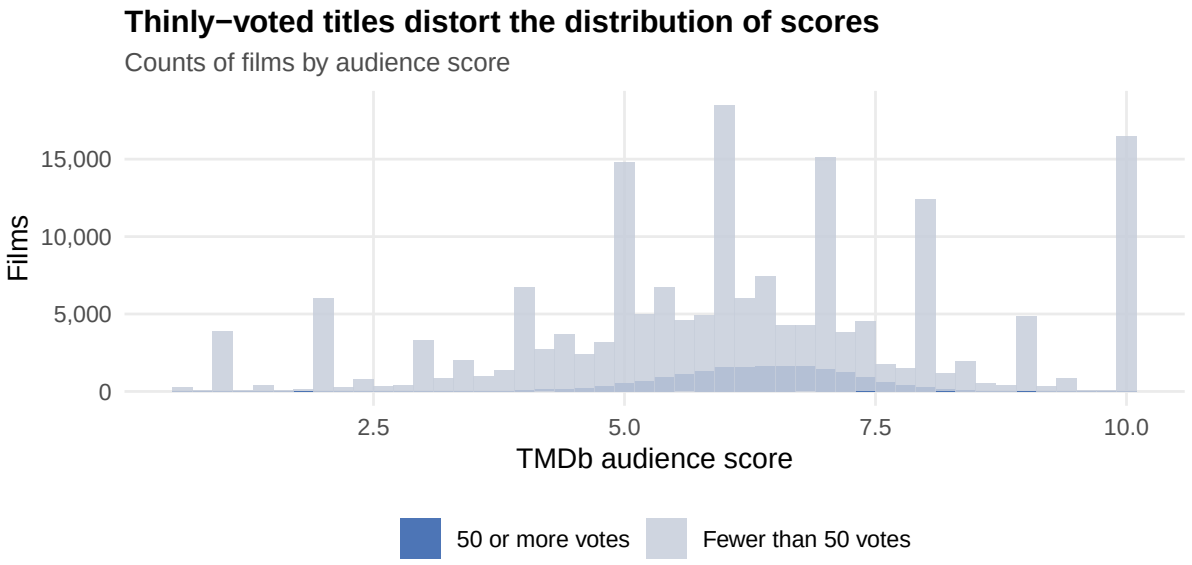


Figure 2: Distribution of audience scores, split by whether the film has at least 50 votes. The spikes at whole and half numbers in the pale series are an artefact of very small vote counts and are the reason for the threshold.

3.3 Data preparation

Preparation is implemented as five ordered scripts in `munge/`, each caching its output so that re-running the project does not repeat work:

Script	Output	Purpose
`01-A-combine.R`	`movies_raw`	Stack the 25 yearly files safely
`02-B-clean.R`	`movies_clean`, `cleaning_audit`	Typing, de-duplication, scoping, sentinel handling, derived variables
`03-C-quality.R`	`data_profile`, `quality_by_year`	Surface properties and completeness over time
`04-D-runtime.R`	`movies_rt`, `band_summary`	Cycle 1 analysis subset and runtime banding
`05-E-genre.R`	`genre_rt`, `genre_summary`	Cycle 2 genre reshape and within-genre statistics

Two preparation decisions deserve spelling out.

The analysis subset. `movies_rt` requires a runtime between 40 and 240 minutes, a recorded score, and at least 50 votes behind it. That leaves 18,329 films, 3.2% of the cleaned table. All three thresholds are named constants at the top of `munge/04-D-runtime.R`, so the sensitivity of the conclusions to any of them can be tested by changing one line.

The banding. Runtime is continuous, but the analysis groups it into five bands (under 80, 80-99, 100-119, 120-149, 150 and over). This is a deliberate choice rather than a convenience. A single

correlation coefficient assumes the relationship has one direction throughout; banding makes no such assumption and lets the shape of the relationship show itself. As Figure 4 demonstrates, that assumption would have been wrong here.

3.4 Analysis

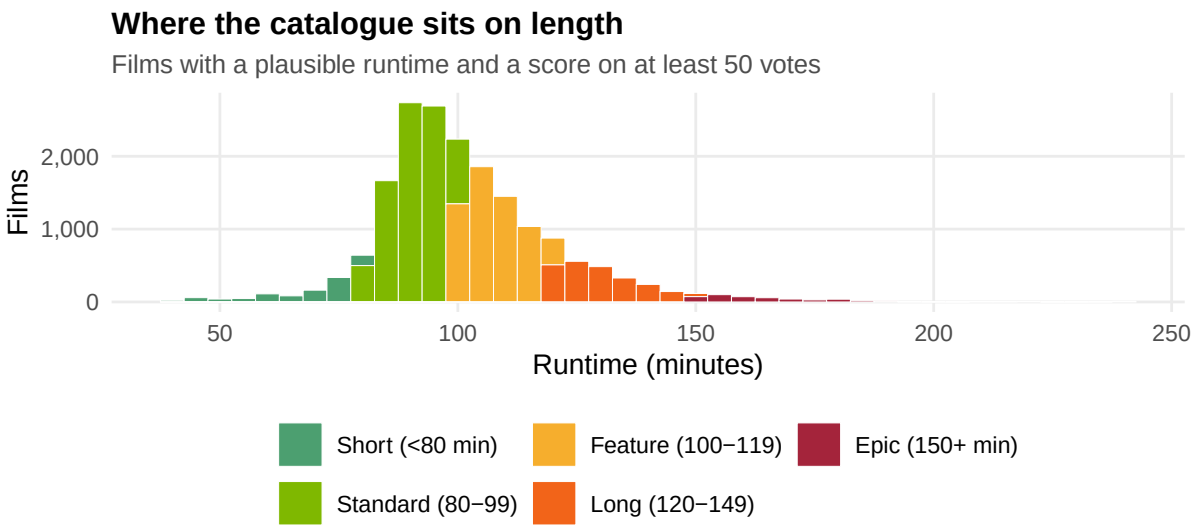


Figure 3: Distribution of runtimes in the analysis subset, coloured by band. The catalogue clusters heavily around 90 minutes, which is where the industry default sits.

Table 3: Audience score by runtime band, analysis subset only.

Runtime band	Films	Median runtime	Mean score	SD of score	Scoring 7 or more
Short (<80 min)	1,005	72	6.65	0.97	37.6%
Standard (80-99)	8,477	91	6.01	0.89	13.9%
Feature (100-119)	6,058	107	6.46	0.77	24.5%
Long (120-149)	2,308	128	6.84	0.70	43.5%
Epic (150+ min)	481	162	7.04	0.77	55.5%

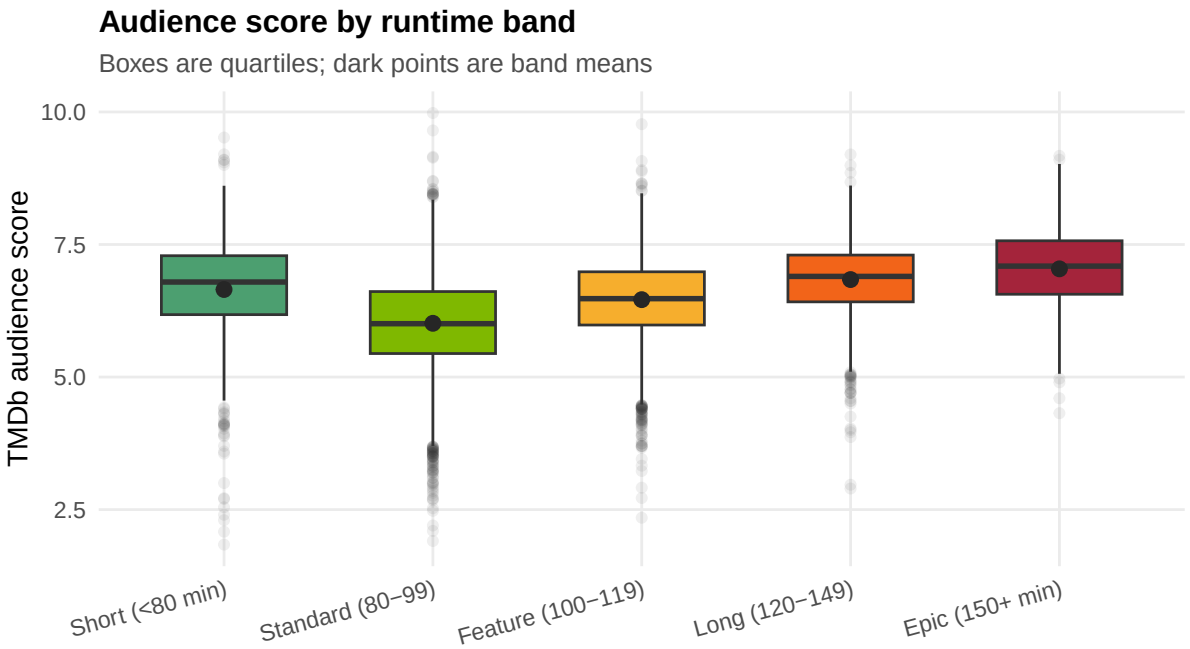


Figure 4: Audience score by runtime band. Boxes show the interquartile range and median; the dark points are band means. The relationship is not a straight line: scores fall from the shortest band to the 80 to 99 minute band, then rise steadily with every step up in length.

3.5 Evaluation of Cycle 1

Against the success criteria set at the start of this cycle:

Criterion 1 — quantify direction and size. Met. Mean audience score ranges from 6.01 in the worst-performing band to 7.04 in the best, a spread of 1.03 points. The share of films scoring 7 or more rises from 13.9% to 55.5% across the same range.

Criterion 2 — establish whether the association is monotonic. Met, and the answer is **no**. This is the substantive finding of the cycle. Scores fall from the under-80 band to the 80-99 band, then rise at every subsequent step. The relationship is U-shaped with its trough at 80 to 99 minutes — which is also the largest band in the catalogue, holding 8,477 of the 18,329 films analysed. A single correlation coefficient, which is the obvious way to approach this question, would have averaged the falling and rising portions together and concealed the shape entirely.

Criterion 3 — materiality threshold of 0.25 points. Met with room to spare. The observed spread of 1.03 points is over four times the threshold.

A second finding, not anticipated by the criteria. The standard deviation of scores *falls* as films get longer, from 0.97 in the shortest band to 0.70 in the 120-149 band. Longer films are not merely better received on average; they are more *consistently* received. For a commissioner, that is a separate and arguably more useful fact than the level, because it speaks to the risk attached to a decision rather than its expected value.

What this cycle cannot settle, and why Cycle 2 is needed. Genres are made at characteristic lengths. Animation runs to a median of 87 minutes and History to 118; documentaries and animated films are among the best-rated genres in the catalogue and among the shortest. It is therefore entirely

possible that Figure 4 shows nothing about length at all, and is instead a picture of which genres happen to occupy which part of the runtime range. The upturn at the short end is particularly suspicious on these grounds.

Next steps. Three follow — the first is taken up as Cycle 2, and the other two are recorded in the Deployment section as candidates for a third cycle:

1. Recompute the runtime-score relationship *within* each genre. If it survives, it is a property of length; if it collapses, it was genre all along.
2. Test whether the effect is stable over the 25-year period, since both audience behaviour and TMDb's coverage changed considerably.
3. Test sensitivity to the 50-vote threshold, which is the strongest assumption in the preparation.

4 Cycle 2: does the relationship survive genre?

4.1 Business understanding, revised

Cycle 1 leaves the commissioning team with a finding they cannot yet act on. A house rule about length is only worth changing if length is doing the work; if the pattern is really about genre, then the correct advice concerns commissioning mix, not runtime.

The revised question is therefore narrower and directly answers the objection:

> Within each genre taken separately, does audience score still rise with runtime — and does the strength of that association vary in a way a commissioner could use?

Success criteria. This cycle will be considered to have answered its question if it can:

1. produce, for every genre with at least 100 films in the subset, a single statistic relating runtime to score that is comparable across genres;
2. conclude that the Cycle 1 relationship is **not** an artefact of genre composition if the median of those within-genre statistics is positive; and
3. identify which genres depart from the general pattern, since an undifferentiated result would not change any commissioning decision.

4.2 Data understanding and preparation

`genres` arrives as a single comma-separated string per film ("Action, Drama, Adventure"), which cannot be grouped on. `expand_genres()` in `lib/helpers.R` reshapes the subset to one row per film-genre pair, so a film with three tags contributes three rows. **Consequence:** every genre figure in this cycle counts *tags*, not films, and totals across genres exceed the number of films. This is stated wherever it matters.

Two further points of preparation. Genres with fewer than 100 films are excluded, because a correlation computed on a few dozen titles is not stable enough to compare against one computed on several thousand. And the statistic used is **Spearman's rank correlation** rather than Pearson's, because neither runtime nor score is symmetrically distributed and rank correlation does not assume they are.

4.3 Analysis

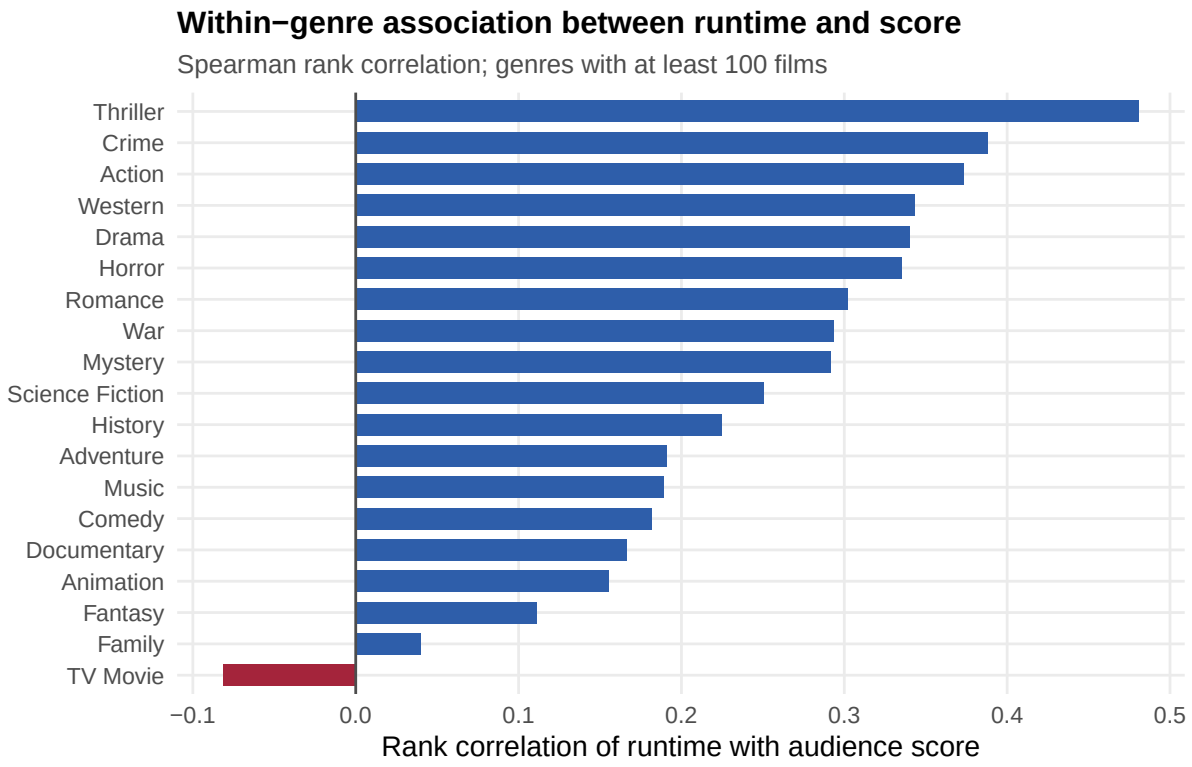


Figure 5: Rank correlation between runtime and audience score, computed separately within each genre. Bars to the right of zero mean longer films score better within that genre. Only one genre falls below zero.

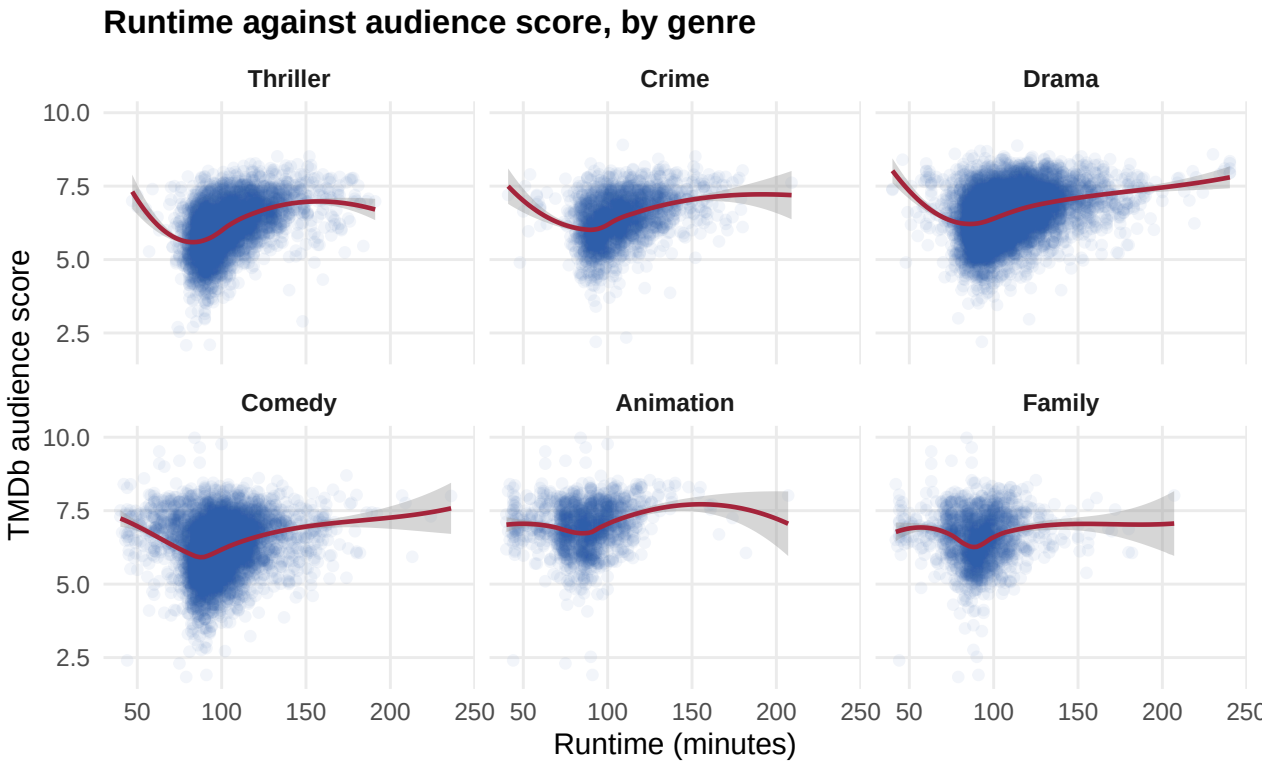


Figure 6: Runtime against audience score for six genres spanning the range of Figure 5, with a smoothed trend. The upward slope is clear in the narrative genres on the top row and close to flat in the three below.

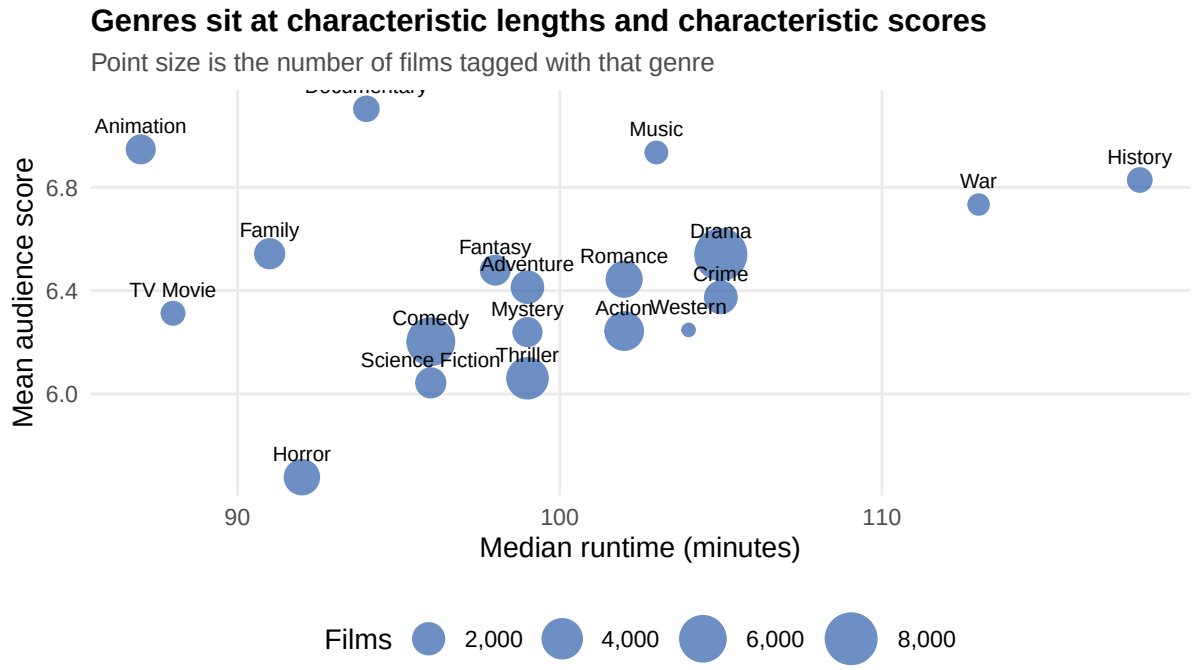


Figure 7: Median runtime against mean score for each genre. The best-rated genres, Documentary and Animation, are also among the shortest, which is what produced the upturn at the short end of Figure 4.

4.4 Evaluation of Cycle 2

Against the success criteria set at the start of this cycle:

Criterion 1 — a comparable statistic per genre. Met. `genre_summary` provides a Spearman rank correlation for each of the 19 genres with at least 100 films, shown in Figure 5.

Criterion 2 — is Cycle 1 an artefact of genre? Met, and the answer is **no**. 18 of 19 genres show a positive association between runtime and score, with a median of 0.25. The relationship therefore holds within genre and is not simply a picture of which genres occupy which part of the runtime range. Figure 6 shows the same thing directly: the slope is visible inside individual genres, not only across them.

Criterion 3 — identify departures from the pattern. Met, and this is where the cycle earns its keep. The correlations do not merely vary; they vary in an interpretable way:

- **Strong** — Thriller (0.48), Crime (0.39), Action (0.37), Drama (0.34).
- **Weak or absent** — Family (0.04), Fantasy (0.11), Animation (0.16), TV Movie (-0.08), the only negative value in the set.

The genres where length matters most are narrative and plot-driven, where additional running time buys character development, subplot and consequence. The genres where it matters least are those whose satisfaction does not depend on accumulated narrative. That is a hypothesis rather than a demonstrated mechanism, but it is consistent across the whole ranking rather than being read off one or two cases.

The Cycle 1 anomaly resolves itself. Figure 7 explains the upturn at the short end that made Cycle 1's U-shape suspicious. Documentary (7.10) and Animation (6.95) are the two best-rated genres in the catalogue and are also among the shortest. The under-80 band scores well because of what is *in* it, not because brevity helps. Within Documentary and Animation themselves, the association with length remains weakly positive.

What this cycle cannot establish. Nothing here is causal. Films are made at the length their material requires, and a director trusted with 150 minutes is usually a director with a track record. The association between length and score may reflect the resources and confidence behind a project rather than duration itself. Distinguishing those requires data this analysis does not have.

4.5 Deployment

For the commissioning team, the analysis converts into three positions:

1. **Abandon the house rule and set length expectations by genre.** The data gives no support for a single default. In Thriller, Crime, Action and Drama, capping at 90 minutes is placing projects at the worst-performing point of the observed range. In Family, Animation and Fantasy, the same cap costs nothing measurable.
2. **Treat the 80 to 99 minute band as a warning rather than a target.** It holds more films than any other and rates worst of all five. A project arriving at that length by default rather than by design is worth a second look.
3. **Use the dispersion result, not just the level.** Longer films score more consistently. Where a commissioner is trying to reduce the chance of a poor outcome rather than maximise the expected one, that is the more relevant number, and it points the same way.

A third cycle would take up the two next steps identified in Cycle 1 and not pursued here: whether the relationship is stable across the 25-year period, and how sensitive the conclusions are to the 50-vote threshold. The most valuable addition, though, would be a budget or production-company variable, to test whether length is standing in for the resources behind a project.

5 Limitations

- **The vote threshold selects on visibility.** Requiring 50 votes restricts the analysis to films enough people watched to rate, and obscurity is not random. The subset is 3.2% of the cleaned table. Conclusions describe films that found an audience, not all films made.
- **Audience scores are self-selected.** People who dislike a film often do not finish it or rate it. This inflates scores generally and may inflate them more for long films, which demand more commitment before a viewer is in a position to rate at all.
- **Genre tags are multi-valued and inconsistently applied**, so genre aggregates count tags rather than films.
- **27% of films in the cleaned table carry no genre tag**, and untagged films are likely to be the more obscure ones, adding a second selection effect on top of the vote threshold.
- **Banding discards information.** Five bands make the shape of the relationship visible but treat an 81-minute film and a 99-minute film as identical. Figure 6 is included partly as a check that the banded result is not an artefact of where the cut-points fall.
- **Association, not causation.** Nothing here shows that lengthening a film would improve its reception.

6 Conclusion

Two CRISP-DM cycles took a practical commissioning question — is a 90-minute default costing us anything? — and produced an answer more specific than the question deserved.

Cycle 1 found that audience scores vary with runtime by 1.03 points across the range, and that the relationship is not a straight line: it falls to a trough at 80 to 99 minutes and rises from there. The banding, chosen precisely because it makes no assumption about shape, is what made that visible. Cycle 1 could not rule out that it was looking at genre rather than length, and said so.

Cycle 2 tested that objection directly and rejected it. The association survives within 18 of 19 genres, with a median rank correlation of 0.25. But the more useful result is the variation: length tracks reception strongly in narrative genres and barely at all elsewhere, which turns a general observation into advice a commissioner can apply differently to different projects.

The recommendation is not that films should be longer. It is that the 90-minute default is the least defensible single answer available, because it sits at the bottom of the observed range and applies one rule to genres the data shows behaving differently.

7 Appendix: project structure and reproduction

```
MAS8505_ProjectTemplate/
  config/global.dcf        ProjectTemplate settings
  data/                    25 raw yearly CSVs (read-only)
  lib/helpers.R            base-R helpers, runtime banding, ggplot2 theme
  munge/01-A-combine.R     stack the yearly files
       02-B-clean.R        typing, de-duplication, scoping, derived vars
       03-C-quality.R      surface properties and completeness profiling
       04-D-runtime.R      Cycle 1 subset and runtime banding
       05-E-genre.R        Cycle 2 genre reshape and correlations
  cache/                   cached preprocessed objects
  docs/data_notes.md       exploratory findings recorded during analysis
  reports/                 this document
  renv.lock                fixed package versions
  README                   how to run the analysis
  GitLog.txt               full commit history
```

To reproduce:

```
renv::restore()                 # install the recorded package versions
library(ProjectTemplate)
load.project()                  # data, libraries, helpers, cached objects
rmarkdown::render("reports/MAS8505_runtime_report.Rmd")
```

```
## R version: R version 4.6.1 (2026-06-24 ucrt)
```

```
## Platform : x86_64-w64-mingw32/x64
```

```
## Key packages: digest, dplyr, ggplot2, knitr, ProjectTemplate, scales, tibble
```